MATHEMATICS DEPARTMENT
FAITH · RIGHTEOUSNESS · WISDOM

cognia SCHOOL OF DISTINCTION 2024

مدارس دار الفكر
DAR ALFIKR SCHOOLS

GRADE 10 · ALGEBRA II · Week 8

Pre-Session

Lesson 2-6

The Quadratic Formula

OBJECTIVES

- Analyze the discriminant to identify the number and type of solutions for a quadratic equation.
- Solve quadratic equations using the quadratic formula, including those with nonreal solutions.

Mr Malek Thiab

Home task before class

Watch · Study · Check · Submit on the LMS before class

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

HSN.CN.C.7 · HSA.REI.B.4.A · HSA.REI.B.4.B

SLIDE 2 · WATCH · LESSON 2-6

The one idea

WATCH BEFORE CLASS

How to use the quadratic formula

Khan Academy · YouTube

https://www.youtube.com/watch?v=i7idZfS8t8w

LISTEN FOR

1. How the three coefficients are read from the equation, with their signs.
2. How the numbers are substituted into the formula and simplified step by step.
3. Why the part under the square root decides how many real solutions there are.

KEY IDEA

$$ax^2 + bx + c = 0 \;\Rightarrow\; x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Delta = b^2 - 4ac$$

$$\Delta > 0 : \text{ two real} \quad \Delta = 0 : \text{ one real} \quad \Delta < 0 : \text{ nonreal}$$

Vocabulary: Discriminant · quadratic formula

THREE THINGS YOU ALREADY KNOW

Squaring a negative

$$(-7)^2 = 49$$

The bracket is squared, so the answer is positive.

Radical fractions

$$\frac{4 + 2\sqrt{5}}{2} = 2 + \sqrt{5}$$

Divide every term of the top by the bottom.

Roots of negatives

$$\sqrt{-36} = 6i$$

From Lesson 2-4: the root of a negative uses the imaginary unit.

SLIDE 3 · STUDY · LESSON 2-6

One worked example

Solve: $2x^2 - 7x + 3 = 0$

Cover the right-hand side, try each step yourself, then check.

1	Write the three coefficients WITH their signs.	$a = 2, \quad b = -7, \quad c = 3$
2	Find the discriminant first.	$b^2 - 4ac = (-7)^2 - 4(2)(3) = 49 - 24 = 25$
3	Substitute. The minus in front of the b turns into a plus.	$x = \frac{-(-7) \pm \sqrt{25}}{2(2)} = \frac{7 \pm 5}{4}$
4	Split into the two solutions.	$x = \frac{12}{4} = 3 \quad \text{or} \quad x = \frac{2}{4} = \frac{1}{2}$
5	Check one solution in the original.	$2(3)^2 - 7(3) + 3 = 18 - 21 + 3 = 0$

The mistake most people make: keeping b instead of its opposite. The numerator then starts with minus seven instead of plus seven, and neither answer passes the check.

SLIDE 4 · CHECK · LESSON 2-6

Four questions: answer before class

This is not graded. It tells me what to teach first, so an honest answer helps you more than a lucky one.

1. Evaluate the discriminant of this equation. $3x^2 - 5x + 1 = 0$

2. Simplify this fraction fully. $\frac{6 + \sqrt{12}}{2}$

3. Write this using i. $\sqrt{-49}$

4. A parabola never crosses the x-axis. Is this positive, zero or negative? $b^2 - 4ac$

HOW TO SUBMIT

- Type your four answers into the LMS form, one line each.
- If you are stuck, write "not sure yet" and say what confused you. That is a useful answer.
- Due before the bell. It takes three minutes.

In class we start with: an 8-question Wayground kick-off (about 4 minutes) on the formula, the discriminant and the four skills above. Then we go straight into the quadratic formula.